METHOD STATEMENT

WATERPROOFING WORKS

SikaTop® Seal-107 (RC Gutter & Scupper Drains) • Sikalastic®-632 R (Balcony Floors)

Project	The Proposed Provision and Laying of Waterproofing Membrane to Existing Roof Balconies and Associated Works at Wisma Bintulu Building
Location	Wisma Bintulu, No. 1, Jalan Tanjung Kidurong, P.O. Box 55, 97007 Bintulu, Sarawak
Client	Lembaga Kemajuan Bintulu (Bintulu Development Authority, BDA)
Contract / Drawing Ref.	BDA-MWS/BLD/1/2025 – Drawings 1/11 to 11/11 (March 2025)
Contractor	Holy Builders Sdn. Bhd.
Document No.	HB/MS/WB-WP/2026-001
Revision	Rev. 0
Date	October 2026

Document Approval

	Name	Signature	Date
Prepared by (Site Engineer)			
Reviewed by (Project Manager)			
Approved by (Director, Holy Builders Sdn. Bhd.)			
Reviewed by (Sika Technical Services)			
Accepted by (Client – BDA)			

Revision History

Rev.	Date	Description	Issued by
0	Oct 2026	First issue for client approval	Holy Builders Sdn. Bhd.

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1. PURPOSE

This Method Statement describes the materials, sequence of work, quality control and safety measures that Holy Builders Sdn. Bhd. will adopt for the provision and laying of waterproofing to the existing roof balconies, perimeter RC gutter and scupper drains at Wisma Bintulu. It is submitted for the Client's review and approval prior to commencement of the works and shall be read together with the contract drawings and the Sika product data sheets listed in Section 3.

2. SCOPE OF WORKS

The works comprise the cleaning, preparation and waterproofing of the areas identified on drawings BDA-MWS/BLD/1/2025 (1/11 to 11/11), summarised below. Areas are plan areas measured from the contract CAD drawing.

Floor	Element	Area (m²)	Waterproofing system
1st Floor	Existing perimeter RC gutter (99.6 m outer dia. x 1.0 m wide) – clean and waterproof	309.84	SikaTop® Seal-107 – 3 coats
3rd – 6th Floor	New 300 mm wide x 50 mm deep scupper drains	337.90	SikaTop® Seal-107 – 3 coats
3rd – 6th Floor	Roof balcony floors – "Proposed area for waterproofing work as per contract"	2,688.10	Sikalastic® U-Primer + Sikalastic®-632 R – 3 layers
3rd – 6th Floor	Associated works: new 100/110 mm dia. uPVC RWDP coring through slab / gutter wall, floor trap covers, sealing of penetrations	–	As per drawings

Ground and 2nd floor plans contain RWDP coring only; no waterproofing areas are shown on these floors.

3. REFERENCE DOCUMENTS

- Contract drawings BDA-MWS/BLD/1/2025, sheets 1/11 (Ground Floor) to 7/11 (6th Floor), 8/11 (Typical Section along Grid), Details A, 11/11 (Typical Cross-Section of Slab), March 2025.
- Sika Product Data Sheet – SikaTop® Seal-107 (MY), 2-part polymer-modified cementitious waterproofing slurry.
- Sika Product Data Sheet – Sikalastic®-632 R, 1-part moisture-triggered polyurethane liquid-applied roof waterproofing membrane (Sika Malaysia, Version 04.01, Oct 2020).
- Sika Product Data Sheet – Sikalastic® U-Primer (Sika Malaysia, Version 03.01, Apr 2023).
- Sika Method Statement – Sikalastic®-632 R Liquid Applied Membrane.
- Sika Safety Data Sheets for all products listed above.
- Occupational Safety and Health Act 1994 (Act 514) and CIDB / DOSH guidelines for working at height.
- Holy Builders Sdn. Bhd. Project Quality Plan and HSE Plan.

4. DEFINITIONS AND ABBREVIATIONS

Term	Meaning
BDA	Bintulu Development Authority (Lembaga Kemajuan Bintulu) – the Client
PDS	Sika Product Data Sheet

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Term	Meaning
RWDP	Rain water down pipe
SSD	Saturated surface dry – substrate damp but without standing water
DFT	Dry film thickness
ITP	Inspection and Test Plan
PPE	Personal protective equipment
HP / WP	Hold point / Witness point in the ITP

5. RESPONSIBILITIES

Role	Responsibility
Project Manager	Overall planning, resources, coordination with the Client and Sika; approval of this Method Statement and the ITP.
Site Engineer / Supervisor	Day-to-day supervision, setting out of areas, substrate inspection, recording of inspections, material batching control, weather checks and coordination of hold points.
Safety & Health Officer / Site Safety Supervisor	Toolbox briefings, permit-to-work for working at height and hot/solvent works, inspection of edge protection and PPE, incident reporting.
Waterproofing Applicator (Sika-trained)	Execution of surface preparation, priming, mixing and application strictly in accordance with the Sika PDS and this Method Statement.
Sika Technical Services	Pre-start training/briefing, site visits at key stages, review of substrate test results, and issue of product warranty on completion.
Client / BDA Representative	Witness of hold points, approval of substrate before coating and acceptance of flood tests.

6. MATERIALS

6.1 Product allocation

The Sika specification allocates the two products as follows. The allocation follows the drawing call-outs on sheet 11/11 ("New 3 layer coat of Sika Top Seal 107 (MY)" at slab / scupper level and "New 3 Layer of Sikalastic-632 R Water proofing Coat" on the balcony floor).

Element	Product	Reason for selection
Existing RC gutter (1st floor)	SikaTop® Seal-107 (MY), 3 coats	Cementitious slurry bonds to the damp, cleaned concrete gutter, is suitable for permanently wet / water-retaining structures (gutters, drains, tanks), resists ponding and does not require a dry substrate or solvent primer.
Scupper drains (3rd – 6th floor)	SikaTop® Seal-107 (MY), 3 coats	Water-retaining drain channel leading to the RWDP outlets; same system as the gutter, lapped by the Sikalastic®-632 R floor membrane with local Sika® Reemat reinforcement at the drain edge.
Balcony floors (3rd – 6th floor)	Sikalastic® U-Primer + Sikalastic®-632 R, 3 layers	Seamless, elastic, crack-bridging and UV-resistant exposed roof membrane suitable for falls and maintenance foot traffic; fully detailed around RWDP outlets, pipe penetrations, upstands and raking columns.

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6.2 Material schedule

Product	Pack	Coats	Consumption (PDS)	Use
SikaTop® Seal-107 (MY)	25 kg set (A 5 kg + B 20 kg)	3	1.5 kg/m² per coat	Gutter and scupper drains
Sikalastic® U-Primer	14 kg pail	1	0.20 kg/m²	Primer on concrete floors
Sikalastic®-632 R	21 kg pail	3	0.70 kg/m² per coat (2.10 kg/m² total)	Balcony floor membrane
Sika® Reemat Premium / Sikalastic® Fleece	Roll	–	As detailed	Local reinforcement at drains, outlets, upstands, cracks
Sika® Reactivation Primer	Pail	–	As PDS	Only if a Sikalastic®-632 R coat is overcoated after more than 2 days
Sika® repair mortar (SikaTop®-122 / Sika MonoTop®)	Bag	–	As required	Repair of spalls, honeycombs, drain profile
Sikaflex® PU sealant	Cartridge	–	As required	Sealing of RWDP penetrations and joints

Material quantities including the 18% wastage allowance are given in Appendix A.

6.3 Delivery, storage and handling

- All materials shall be delivered in original sealed Sika packaging with batch numbers and shelf life legible. Delivery notes shall be retained for the quality records.
- Materials shall be stored in a dry, shaded, ventilated store on pallets at +5 °C to +30 °C, protected from direct sunlight and rain. Sikalastic®-632 R and Sikalastic® U-Primer are moisture-sensitive: pails shall remain sealed until use and part-used pails shall be re-sealed immediately.
- Shelf life: Sikalastic®-632 R and U-Primer 9 months; SikaTop® Seal-107 12 months. Stock shall be rotated on a first-in first-out basis and expired material removed from site.
- Solvent-based products shall be stored away from ignition sources with SDS and a dry-powder extinguisher at the store.

7. PLANT, TOOLS AND EQUIPMENT

Category	Items
Access & protection	Edge protection / guard rails, safety harness and anchor points, mobile scaffold / tower for raking columns, barricades, warning signage, protective sheeting for adjacent finishes and glazing
Surface preparation	High-pressure water jet (min. 150 bar), diamond grinder with dust extraction, wire brushes, scrapers, industrial vacuum, blower, chipping hammer, hammer-test hammer
Coring & drainage	100/110 mm diamond core drill, uPVC pipes and fittings, new floor trap covers

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Category	Items
Mixing	Slow-speed electric mixer (300–400 rpm) with helical paddle, clean mixing pails, weighing scale, measuring jugs, clean water supply
Application	Stiff synthetic-bristle brushes and brooms (SikaTop® Seal-107), short-pile solvent-resistant rollers, brushes and notched spreaders (Sikalastic®-632 R), scissors for Reemat, sealant guns
Testing & inspection	Moisture meter, pull-off adhesion tester, wet-film thickness gauge / comb, thermometer and hygrometer, dew-point calculator, camera for records
Clean-up	Sika® Thinner C / Colma Cleaner for tools, waste drums, rags, spill kit

8. PRE-COMMENCEMENT ACTIVITIES

- Approval of this Method Statement, the ITP and the material submittals (PDS, SDS, samples, Sika warranty undertaking) by the Client.
- Joint site inspection with the Client to confirm the extent of the gutter, drains and floor areas, locations of new RWDP coring and any existing defects (cracks, spalls, ponding, blocked outlets) which are recorded on marked-up plans with photographs.
- Sika Technical Services pre-start briefing to the applicator team; applicators shall be Sika-trained and the names recorded.
- Permit-to-work for working at height, hot works (if any) and solvent handling; toolbox briefing on this Method Statement.
- Install edge protection, barricade the work zone and protect the building façade, glazing, aluminium frames and floor finishes below the work areas. Divert building users away from the balconies and gutter during the works.
- Trial area of approximately 10 m² for each system (gutter and floor) to confirm consumption, appearance and adhesion; the trial area is inspected with Sika and the Client before full application (Hold Point).

9. METHOD OF WORKS

9.1 General surface preparation (all areas)

- Remove all debris, vegetation, silt, existing failed coatings, loose screed and laitance by scraping, grinding and high-pressure water jetting. Blocked RWDP outlets are rodded and flushed clear.
- Hammer-test the whole surface including upstands. Hollow or delaminated screed/concrete is broken out to sound material and reinstated with Sika® repair mortar to the manufacturer's instructions; cracks wider than 0.3 mm are routed and sealed with Sikaflex® PU sealant.
- Form falls to outlets where ponding is identified (min. 1:80) using Sika® repair mortar; the 300 mm x 50 mm scupper drain profile is re-formed where damaged so that it drains to the RWDP.
- Complete the new 100/110 mm uPVC RWDP coring through the slab / gutter wall as shown on the drawings before any coating. Core holes are made with a diamond core drill, pipes fixed with sleeves, and the annulus sealed with non-shrink grout and Sikaflex® PU sealant. Existing floor trap covers are replaced with new approved covers.
- Blow and vacuum the surface free of dust. The prepared substrate is inspected and signed off by the Site Engineer and the Client before coating (Hold Point).

9.2 Existing RC gutter – SikaTop® Seal-107 (3 coats)

Step 1. Clean the gutter completely as in 9.1 and check that the gutter falls freely to the outlets. Repair all defects and allow repair mortar to cure.

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Step 2. Pre-wet the concrete thoroughly to a saturated surface dry (SSD) condition. Remove any standing water; the surface must be damp but not glistening.

Step 3. Mixing: pour Part A (liquid, 5 kg) into a clean pail, add Part B (powder, 20 kg) gradually while mixing with a slow-speed mixer for 3 minutes until a lump-free slurry is obtained (A : B = 1 : 4 by weight). Mix only full sets; do not add water. Pot life is approximately 30–40 minutes at 30 °C – mix only what can be applied in that time.

Step 4. 1st coat: apply by stiff brush at approximately 1.5 kg/m², working the slurry firmly into the surface in one direction, including the gutter base, both side walls to the top of the gutter and 100 mm onto the slab, and around every outlet. Fill all pinholes.

Step 5. 2nd coat: once the first coat has hardened (approximately 4–8 hours at > 20 °C; maximum 48 hours; lightly re-wet with a fine mist if longer than 12 hours), apply the second coat at approximately 1.5 kg/m² crosswise to the first.

Step 6. 3rd coat: apply the third coat as specified on the drawing in the same manner, again crosswise, to give a total of approximately 4.5 kg/m² and a minimum finished thickness of 2 mm. Do not exceed 2 kg/m² in any single coat.

Step 7. At RWDP outlets, dress the slurry 50 mm into the pipe mouth; embed Sika® Reemat where cracks or construction joints cross the gutter.

Step 8. Curing: protect the fresh coating from rain, direct sun and strong wind for at least 24 hours and keep it damp (fine mist / wet hessian) for 3 to 5 days to ensure full hydration and to prevent shrinkage cracking. Allow a minimum of 2 days air curing before any water exposure and 7 days before flood testing.

Step 9. Flood test the gutter (bung outlets, 50 mm head, 24 hours) in the presence of the Client; inspect soffits for leakage, record and rectify any defect, then re-test (Witness Point).

9.3 Scupper drains (3rd – 6th floor) – SikaTop® Seal-107 (3 coats)

Step 1. Set out the 300 mm wide x 50 mm deep scupper drain ring on each floor as drawn, re-form the channel with Sika® repair mortar where required and ensure a continuous fall to each new RWDP floor trap. Allow repair mortar to cure (min. 3 days).

Step 2. Pre-wet the channel to SSD and apply SikaTop® Seal-107 in three coats exactly as described in 9.2 to the drain base, both sides of the channel and 100 mm onto the adjacent floor on each side, dressing into each outlet. Total approximately 4.5 kg/m².

Step 3. Fit the new approved floor trap covers after the final coat has cured.

Step 4. The 100 mm lap onto the floor is later over-coated by the Sikalastic®-632 R floor membrane (Section 9.4) so that the two systems are continuous. The cementitious coat must be fully cured (min. 7 days) and dry before priming.

Step 5. Flood test each drain ring for 24 hours and check all outlets before the floor membrane is applied (Witness Point).

9.4 Balcony floors (3rd – 6th floor) – Sikalastic® U-Primer and Sikalastic®-632 R (3 layers)

Substrate acceptance

- Concrete / screed must be at least 28 days old, sound, clean and dry: moisture content < 4% by weight (moisture meter, min. 3 readings per 100 m²), pull-off adhesion ≥ 1.5 N/mm², surface profile CSP 2–3 after grinding, free of laitance, oil, dust and loose material.
- Environmental limits during priming and coating: ambient and substrate temperature +10 °C to +40 °C, relative humidity ≤ 85%, substrate temperature at least 3 °C above dew point, no rain expected within the skin-forming time. Readings are recorded on the daily application record.

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Application sequence

Step 1. Mask and protect the completed scupper drains, floor trap covers, glazing, aluminium frames and column bases. Mark out daily work areas so that each coat terminates at a neat line.

Step 2. Priming: stir Sikalastic® U-Primer and apply by short-pile roller or brush in one thin, even coat at approximately 0.20 kg/m² to the whole floor area, including the 100 mm lap over the SikaTop® Seal-107 drain edge, upstands and around outlets. Avoid puddles. Allow to become tack-free (approximately 2–4 hours; overcoat within 24 hours, otherwise re-prime).

Step 3. Detailing (before the main coats): at the scupper drain edges, RWDP outlets, pipe penetrations, upstands, raking column bases, cracks and construction joints, apply a stripe coat of Sikalastic®-632 R approximately 300 mm wide, embed Sika® Reemat Premium / Sikalastic® Fleece into the wet coat with a brush so that it is fully wetted-out without air pockets, and apply a further coat over the reinforcement. Fit Sikalastic® outlet reinforcement / flashing pieces at each RWDP outlet and dress the membrane 50 mm into the outlet.

Step 4. 1st layer: stir Sikalastic®-632 R thoroughly without entraining air and apply by roller / brush at approximately 0.70 kg/m², spreading evenly and working into the surface. Check wet film thickness with a comb gauge (approximately 500 µm).

Step 5. 2nd layer: once the first layer is tack-free and can be walked on (typically 4–8 hours at 30 °C / 50% RH, depending on humidity), apply the second layer at approximately 0.70 kg/m² at right angles to the first. Overcoat within 2 days; if more than 2 days elapse, clean the surface and apply Sika® Reactivation Primer before continuing.

Step 6. 3rd layer: apply the third layer in the same manner to give a total of approximately 2.10 kg/m² (minimum 1.5 mm DFT). The finished membrane shall be continuous, pinhole-free, of uniform colour and terminated with the detailing at drains, outlets and upstands.

Step 7. Lap each day's work a minimum of 100 mm onto the previous day's membrane and stagger the termination lines of successive layers.

Step 8. Curing: rain-resistant after approximately 1 hour; foot traffic after 24 hours; full cure 3–5 days. Protect the finished membrane from mechanical damage, other trades and sharp objects. No further works are permitted on the membrane without protection boards.

Step 9. On completion of each floor, inspect the membrane with the Client and Sika, carry out a 24-hour flood test on the balcony (outlets bunged, maximum 25 mm head), inspect the soffit below, rectify and re-test if required, then release the area (Witness Point / Hold Point).

9.5 Sequence and programme

- Work proceeds floor by floor from the 6th floor downwards so that completed lower areas are not damaged by works above, with the 1st floor gutter programmed in parallel.
- Each floor is divided into working bays (typically two to four bays per ring) so that cleaning, repair, priming and coating follow in sequence and freshly coated areas are not trafficked.
- Coating is carried out in the morning after confirming the weather forecast; application stops if rain threatens and uncured SikaTop® Seal-107 is covered.

10. QUALITY CONTROL AND INSPECTION

The Inspection and Test Plan below forms part of this Method Statement. Records (checklists, photographs, batch numbers, environmental readings, consumption per bay and flood test reports) are compiled into the final handover file together with the Sika warranty.

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No.	Activity	Acceptance criteria	Type	By
1	Material delivery	Sika packaging, batch no., within shelf life, matches submittal	Review	Site Engineer
2	Trial area (gutter and floor)	Adhesion, appearance and consumption acceptable to Sika / Client	HP	Client / Sika / HB
3	Substrate after preparation (each bay)	Sound, clean, hammer-test OK, repairs cured, falls to outlets, RWDP coring complete	HP	Client / HB
4	Moisture and environmental readings (floors)	Moisture < 4%, temp +10 to +40 °C, > 3 °C above dew point, RH ≤ 85%	Record	Site Engineer
5	Pull-off test (floors)	≥ 1.5 N/mm², 1 test per 500 m² minimum	Record	Site Engineer / Sika
6	Primer application	Even coat, no puddles, ~0.20 kg/m², overcoated within 24 h	Surveillance	Site Engineer
7	Detailing and reinforcement	Reemat fully wetted-out, outlets dressed, no voids	WP	Client / HB
8	Each coat – SikaTop® Seal-107 and Sikalastic®-632 R	Consumption per bay within ±10% of specified, WFT checked, no pinholes / holidays, overcoating times respected	Record / WP	Site Engineer
9	Final membrane inspection	Continuous, uniform, laps and terminations correct, total consumption achieved	WP	Client / Sika / HB
10	Flood test (gutter, drains, each balcony)	24 h, no leakage to soffit, outlets free-flowing	HP	Client / HB
11	Handover	As-built marked-up plans, QC records, Sika warranty issued	Review	Client / HB

11. HEALTH, SAFETY AND ENVIRONMENT

- Working at height: the balconies and gutter are at the building perimeter. Edge protection shall be installed before work starts; where it cannot be provided, workers shall wear full-body harnesses attached to certified anchor points. A permit-to-work is required daily.
- Chemical safety: Sikalastic®-632 R and Sikalastic® U-Primer are solvent-based and flammable; SikaTop® Seal-107 Part B is alkaline. Workers shall wear nitrile gloves, safety goggles, long sleeves, safety boots and, in enclosed or poorly ventilated areas, organic-vapour respirators. No smoking, naked flames or hot works within 10 m of the application area or the store. SDS are kept on site and briefed at the toolbox talk.
- Dust and noise: grinding and coring are carried out with dust extraction and water suppression; ear protection and dust masks are worn. Working hours comply with the Client's building operation requirements.

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- Manual handling: 25 kg sets and 21 kg pails are lifted by two persons or mechanically; materials are hoisted to the floors in secured cages.
- Weather: works stop during rain, lightning or strong wind. Uncured material is protected.
- Housekeeping and environment: waste, empty pails and rags are segregated and disposed of through a licensed contractor; no washing of tools into the RWDP or drains; spill kits are available; the work area is cleaned daily.
- Emergency: first-aid kit, trained first-aider, emergency contact numbers and the nearest clinic/hospital route are displayed at the site office.

12. CURING, PROTECTION AND HANDOVER

- Completed areas are barricaded until fully cured (SikaTop® Seal-107: 7 days; Sikalastic®-632 R: 3–5 days) and are not used as access routes by other trades.
- On completion, all protection is removed, outlets and floor traps are checked clear, and the areas are cleaned and jointly inspected.
- The handover file contains: approved Method Statement and ITP, material delivery records, daily application records with batch numbers and consumption, inspection checklists, flood test reports, photographs, as-built marked-up drawings and the Sika product warranty together with the Holy Builders Sdn. Bhd. workmanship warranty.

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APPENDIX A – MATERIAL QUANTITIES (INCLUDING 18% WASTAGE)

Areas are measured from the contract CAD drawing (plan areas). Consumption rates are the Sika PDS theoretical values; an 18% top-up is added for wastage, substrate absorption and overlaps at details. Full workings are provided in the accompanying Excel file "Wisma_Bintulu_Sika_Quantity_Takeoff_18pct_Wastage.xlsx".

Product	Element	Area (m²)	Coats	Net (kg)	+18% (kg)	Packs
SikaTop® Seal-107 (MY) – 25 kg set	Existing RC gutter, 1st floor	309.84	3	1,394.28	1,645.25	66
SikaTop® Seal-107 (MY) – 25 kg set	Scupper drains, 3rd–6th floor	337.90	3	1,520.55	1,794.25	72
SikaTop® Seal-107 (MY) – total	Gutter + drains (order as one lot)	647.74	3	2,914.83	3,439.50	138
Sikalastic® U-Primer – 14 kg pail	Balcony floors, 3rd–6th floor	2,688.10	1	537.62	634.39	46
Sikalastic®-632 R – 21 kg pail	Balcony floors, 3rd–6th floor	2,688.10	3	5,645.01	6,661.11	318
Sika® Reemat Premium	Detailing at drains, outlets, upstands	As detailed	–	–	–	TBC

Basis: SikaTop® Seal-107 at 1.5 kg/m² per coat x 3 coats; Sikalastic® U-Primer at 0.20 kg/m²; Sikalastic®-632 R at 0.70 kg/m² per coat x 3 layers (2.10 kg/m² total). Gutter side walls, upstands and drain side faces are not included in the plan areas and will be measured on site.

APPENDIX B – AREA SCHEDULE BY FLOOR

Floor	Drawing	Floor area – Sikalastic®-632 R (m²)	Gutter / drain – SikaTop® Seal-107 (m²)	Remarks
1st Floor	2/11	–	309.84	RC gutter
3rd Floor	4/11	960.85	110.49	Scupper drain
4th Floor	5/11	793.02	96.79	Scupper drain
5th Floor	6/11	553.47	72.02	Scupper drain
6th Floor	7/11	380.76	58.60	Scupper drain
TOTAL		2,688.10	647.74	

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